SaaS Valuation Engine · Quantitative Research
Backtest note

18 US software companies
1 November 2026

POINT-IN-TIME BACKTEST · 2023-03-31 TO 2026-09-30

Cheap for its growth has paid, but the evidence is three years deep

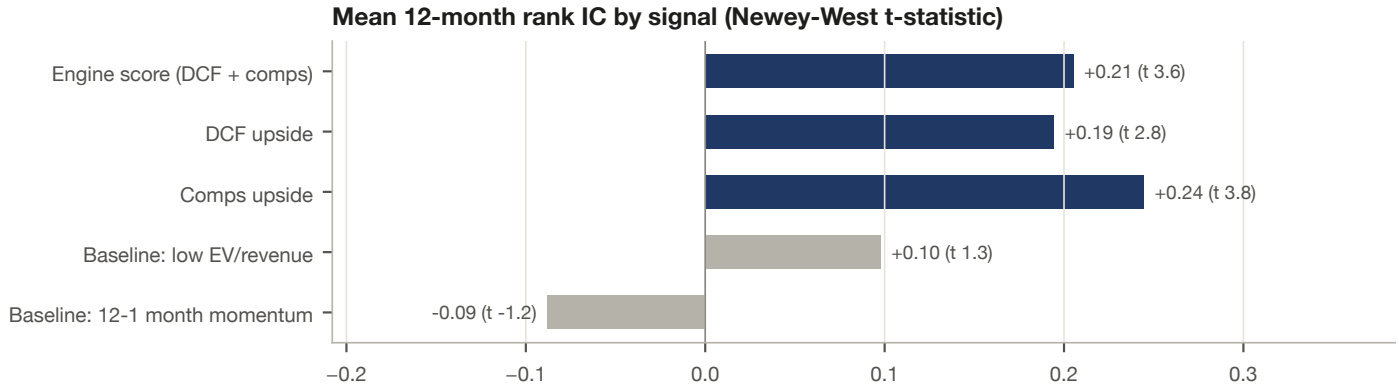
Ranking the 18 companies each quarter by the engine's valuation score, using only filings and prices public on the day, gave a mean 12-month rank information coefficient (IC) of 0.21 (Newey-West t 3.6), positive on 91% of 11 start dates. The cheapest third beat the most expensive third by 15 points over the following year on average. The result survives the checks below, but rests on only three non-overlapping years in one market regime, and should be read as encouraging rather than proven.

Question and method

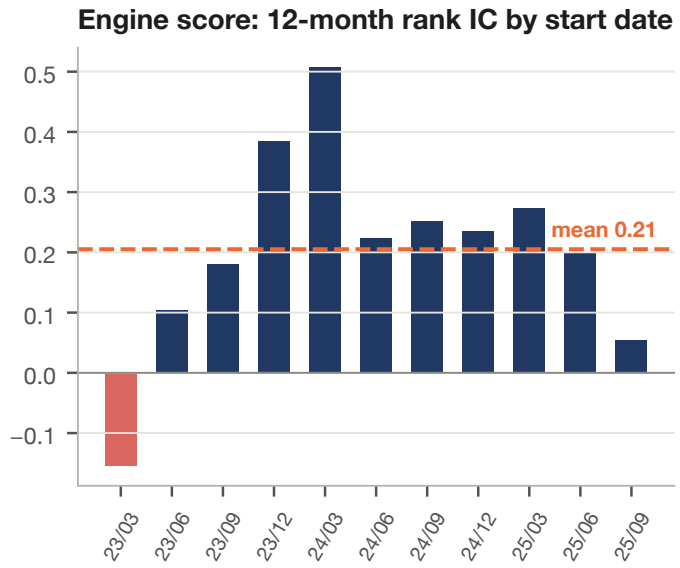
Question. On past dates, did the companies the engine called cheap go on to beat the ones it called expensive?

Point in time. For each quarter-end from 2023-03-31, every SEC fact filed after that day is removed and the engine's own extraction rebuilds the financials as they stood then, including restatements. Prices are that day's closes, betas are regressed on SPY over the prior two years, and the risk-free rate is that day's 10-year Treasury yield. The engine's comps, DCF and ranking then run unchanged. Forward returns are dividend-adjusted.

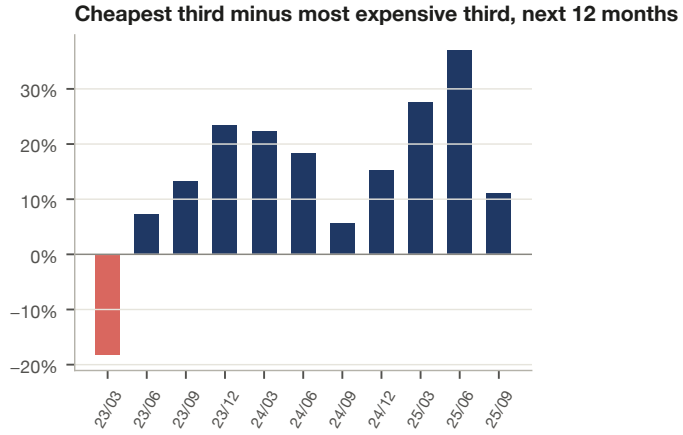
Measures. The rank IC is the Spearman correlation between a signal and the next 12 months' returns across companies on one date. The tercile spread is the average return of the top third minus the bottom third. Start dates are quarterly but returns run 12 months, so windows overlap: t-statistics use Newey-West standard errors.



Grey: naive baselines. Source: this backtest; 12-month horizon, 2023-03-31 to 2025-09-30 start dates.



Orange: mean across start dates.



Red bars: the expensive third did better.

Results

Signal	Mean IC 12m	t (NW)	IC > 0	Tercile spread 12m	Mean IC 6m
Engine score (DCF + comps)	+0.21	3.6	91%	+15 pts	+0.08
DCF upside	+0.19	2.8	91%	+18 pts	+0.09
Comps upside	+0.24	3.8	91%	+17 pts	+0.08
Baseline: low EV/revenue	+0.10	1.3	73%	+5 pts	+0.05
Baseline: 12-1 month momentum	-0.09	-1.2	45%	-8 pts	+0.02

The signal is weaker at six months (+0.08), consistent with valuation gaps closing slowly. Plain cheapness (low EV/revenue) is 80% rank-correlated with the engine score yet earns an IC of only 0.10: the value is in adjusting price for growth, not in low multiples as such. Momentum carries no information here.

Robustness

- **Independent years.** Using one start date per year removes the overlap: ICs of -0.15 (2023), +0.51 (2024), +0.27 (2025). One of three years was negative.
- **No single company.** Dropping each company in turn leaves the mean IC between 0.15 (without TWLO) and 0.25.
- **Chance.** Shuffling the signal across companies on each date 5,000 times, 0.5% of shuffles matched the observed mean IC (95th percentile of the null: 0.12). This null still treats overlapping windows as more independent than they are, so the true p-value is higher.
- **Design chosen after the fact.** The comps regression on revenue growth was selected in 2026 for its fit. Rerun with the rejected alternative (Rule of 40), the score's IC is 0.24 (t 3.3), so the result does not depend on that choice.

Limitations

- **Survivorship.** The 18 companies were chosen in 2026 as today's established SaaS names. Companies that failed, were acquired or shrank out of relevance in 2023-25 are missing, which can flatter any signal.
- **Small, correlated sample.** 18 companies in one sector move together, so the cross-section holds far less independent information than its size suggests, and terciles are six stocks each.

- **One regime.** 2023-26 combined rising-then-stable rates and an AI-driven rotation within software. A signal that rewards cheap-for-growth may behave differently in another cycle.
- **Other design choices.** DCF parameters (terminal growth, mature margin, fade length) were set with today's knowledge. Only the comps choice was tested for sensitivity here.
- **No costs or capacity.** Returns are gross, equal-weighted and assume trading at the close on each date.

Educational project, not investment advice. Data: SEC EDGAR companyfacts (all filed versions, filtered by filing date) and Yahoo Finance prices. Reproduce with `python -m saas_valuation.research.backtest`.